Mixed monopole pairings and a pinched strictly almost-Kähler Einstein example

Qiushi first scientific line: mathematical continuation

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Abstract

Starting from the project's finite-density endpoint proposition, we derive a quantitative two-orientation inequality that allows a positive class to belong to a convex hull of actual monopole classes and explicitly records loss in the mixed intersection pairing. Surface-bundle and symplectic-pair consequences follow from the same statement. Separately, the research system's complete strictly almost-Kähler Einstein metric on $\mathbb{R}^4$ is shown to have sharp global sectional-curvature range $(-2, -1/3]$ in its normalization $\mathrm{Ric} = -3g$. Its smallest curvature-operator eigenvalue is an intrinsic radial coordinate. Every smooth isometric quotient has infinite volume; consequently its Einstein germ cannot occur in a complete finite-volume Einstein manifold. The first part is an implication from the stated FD2 input. The second part uses the supplied complete model and exact curvature blocks, not FD2.

1 Inputs and conventions

Let M be a closed oriented four-manifold, $e = \chi(M) > 0$, and choose the orientation so that $\tau = \mathrm{sign}(M) \geq 0$. Put

$$T_\pm = 2e \pm 3\tau, \quad P = \int |\,\mathrm{Ric}^\circ\,|^2, \quad S = \int s^2, \quad R = P + S/4, \quad \mathcal{J} = 4P/R.$$

Two-forms have their form norm, and Weyl curvature has its curvature-operator Hilbert–Schmidt norm. Thus

$$4\pi^2 T_\pm = S/24 + 2\int |W^\pm|^2 - P/2. \tag{1.1}$$

All class squares and intersections, unless expressly marked otherwise, are evaluated on $[M]$. In particular $b^2[\bar{M}] = -b^2[M]$.

An actual monopole class means the real first Chern class of a fixed Spin^c structure whose unperturbed Seiberg–Witten equations have a solution for every smooth metric in question. Such a class c satisfies the scalar estimate

$$\int s^2 \geq 32\pi^2 \|c^+\|^2. \tag{1.2}$$

A class on $\bar{M}$ satisfies the analogous estimate with the original negative Hodge projection. Ordinary, Bauer–Furuta or covering constructions enter below only when they provide these actual classes; a formal intersection vector is not a substitute for this input. Convex-hull use of actual classes is classical [5].

Input 1.1 (FD2: the project endpoint proposition)**.** On a closed oriented four-manifold, suppose a negative Yamabe representative, normalized by $s = -4$, has the actual conformal monopole supply specified in the project. For an actual class c, assume

$$p_c = \|c^+\|^2 \geq A_c > 0, \qquad c^2 \geq T_+, \qquad \chi/A_c \leq 2.$$

There is one constant $c_{\mathrm{FD}}(2) > 0$, independent of the manifold and its geometry, such that for every finite $\beta \geq 2$,

$$\frac{P + D_\beta(c)}{V} \geq c_{\mathrm{FD}}(2)(1 - 2/\beta)^2, \quad D_\beta(c) = (1 + \beta^{-1})S + 6(\beta + 1)\int |W^+|^2 - 72\pi^2 p_c. \tag{1.3}$$

This is the only nonclassical analytic input to Sections 2–3. It is the $B = 2$ instance used in the supplied two-orientation proof [1, 2]. Set

$$c = \min\{1, c_{\mathrm{FD}}(2)\}, \qquad \varepsilon_{\mathrm{pair}} = (c/200)^2. \tag{1.4}$$

2 A mixed-pairing inequality beyond surface bundles

Theorem 2.1 (Pairing loss and quantitative separation)**.** *Let $\mathcal{C}_+$ be a nonempty finite set of actual monopole classes on M, and let*

$$a \in \mathrm{Hull}(\mathcal{C}_+), \qquad a^2 \geq T_+.$$

Let b be an actual monopole class on $\bar{M}$ with $-b^2 \geq T_-$. Under Input 1.1, every smooth metric satisfies

$$\boxed{\mathcal{J}(g) + (1 - t)_+ + (t - z)_+ \geq \varepsilon_{\mathrm{pair}}, \qquad t = \frac{3\tau}{e}, \quad z = \frac{|a \cdot b|}{e}.} \tag{2.1}$$

The vector a need not itself be a monopole class. No fibration, symplectic splitting or equality of the two class squares is assumed.

Proof. The positivity $a^2 \geq T_+ > 0$ excludes a nonnegative Yamabe representative. Indeed, on such a representative the unperturbed Weitzenböck identity makes every positive projection from $\mathcal{C}_+$ vanish. Then $a^+ = 0$, contrary to $a^2 > 0$. All conformal classes are therefore negative and $R > 0$.

Write $a = \sum_i \lambda_i a_i$, with $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$. By (1.2) and convexity,

$$\|a^+\| \leq \sum_i \lambda_i \|a_i^+\| \leq \sqrt{S/(32\pi^2)}. \tag{2.2}$$

There is no assertion that a Seiberg–Witten solution exists for the convex combination.

If $\mathcal{J}(g) \geq 1$, the conclusion is immediate. Otherwise pass to $\widehat{g} \in [g]$ with $s = -4$. Put $\delta_S = S(g) - S(\widehat{g}) \geq 0$. Equation (1.1) and conformal invariance of Weyl energy give

$$P(g) = P(\widehat{g}) + \delta_S/12, \qquad R(g) = R(\widehat{g}) + \delta_S/3.$$

Consequently $\mathcal{J}(g)$ is a weighted average of $\mathcal{J}(\widehat{g})$ and 1, so

$$j := \mathcal{J}(\widehat{g}) \leq \mathcal{J}(g) < 1.$$

It suffices to prove the inequality for j. All subsequent norms and integrals use $\widehat{g}$.

Set

$$u = \frac{V}{2\pi^2 e}, \quad r = \frac{P}{V} = \frac{4j}{4 - j}, \quad w = \frac{1}{V}\int |W^-|^2, \quad U = \frac{\|a^+\|^2}{e}, \quad Z = \frac{\|b^-\|^2}{e}.$$

Equation (2.2) and the square of a imply $2 + t \leq U \leq u$. Equation (1.1) gives

$$w = \frac{2 - t}{u} - \frac{1}{3} + \frac{j}{4 - j} \geq 0. \tag{2.3}$$

Since $j < 1$, this forces $t < 2$ and

$$2 + t \le u \le C(j)(2 - t), \qquad C(j) = \frac{3(4-j)}{4(1-j)}, \qquad t \le 1 + \frac{9j}{16 - 7j} \le 1 + j. \tag{2.4}$$

The actual scalar estimate for b gives $2 - t \le Z \le u$.

Let

$$d = j + (1 - t)_+ + (t - z)_+.$$

If $d \ge 1/256$, then $d \ge \varepsilon_{\mathrm{pair}}$. Assume $0 < d < 1/256$. Put

$$X = \|a^-\|^2/e = U - a^2/e, \qquad Y = \|b^+\|^2/e = Z + b^2/e.$$

Both are nonnegative, and the class-square assumptions give

$$X \le U - (2 + t), \qquad Y \le Z - (2 - t).$$

Writing $d_- = (1 - t)_+$, estimates (2.3)–(2.4) yield

$$1 - d \le t \le 1 + d, \quad 3 - d \le u \le 3 + 5d < 4, \quad X \le 4d, \quad w \le d, \quad r \le 4d/3. \tag{2.5}$$

For example $C(j) \le 3+3j$ for $j \le 1/4$; hence $X \le 4d_- + 3j(1+d_-) \le 4d$ and $w \le d_-/2 + j/3 \le d$. Crucially,

$$z \ge t - (t - z)_+ \ge 1 - d_- - (t - z)_+ \ge 1 - d. \tag{2.6}$$

The original surface-bundle proof used $z = t$; (2.6) is its replacement.

The same-metric Hodge decomposition and Cauchy–Schwarz give

$$z \le \sqrt{UY} + \sqrt{XZ} \le \sqrt{u\{Z - (2 - t)\}} + 4\sqrt{d}. \tag{2.7}$$

The quantity $1 - d - 4\sqrt{d}$ is positive. Therefore

$$\frac{Z}{u} \ge \frac{2 - t}{u} + \frac{(1 - d - 4\sqrt{d})^2}{u^2} \tag{2.8}$$

$$\ge \frac{1}{3} - \frac{d}{3} + \frac{(1 - d - 4\sqrt{d})^2}{(3 + 5d)^2} \ge \frac{4}{9} - 2\sqrt{d}. \tag{2.9}$$

For the final elementary bound, use $(1-d-4\sqrt{d})^2 \ge 1-2d-8\sqrt{d}$ and $(3+5d)^{-2} \ge 1/9-10d/27$. Their nonnegative product gives the lower bound $1/9 - 8\sqrt{d}/9 - 16d/27$ for the last fraction; adding the first two terms proves the asserted estimate on $d \le 1/256$.

Apply FD2 on $\bar{M}$ to the actual class b, with

$$A_- = T_- = e(2 - t) > 0.$$

Indeed $\|b^-\|^2 \ge -b^2 \ge A_-$, $b^2[\bar{M}] \ge T_+(\bar{M})$, and $e/A_- = 1/(2-t) \le 1/(1-d) < 2$. Its left-hand side is exactly

$$\bar{\kappa}_\beta = r + 16(1 + \beta^{-1}) + 6(\beta + 1)w - 36Z/u.$$

Take the finite parameter $\beta = d^{-1/2} \ge 16$. By (2.5) and (2.9),

$$\bar{\kappa}_\beta \le 94\sqrt{d} + \frac{22}{3}d < 100\sqrt{d}, \qquad \bar{\kappa}_\beta \ge c(1 - 2\sqrt{d})^2 \ge c/2.$$

Thus $d \ge (c/200)^2$.

If $d = 0$, then $j = 0$, $t = 1$, $z \ge 1$, $u = U = 3$, and $X = w = r = 0$. Equation (2.7) gives $Z/u \ge 4/9$, so $\bar{\kappa}_\beta \le 16/\beta$ for every finite β. FD2 contradicts this as $\beta \to \infty$. This completes all cases. Returning from j to $\mathcal{J}(g)$ proves (2.1). □

Corollary 2.2 (A finite-configuration form)**.** *Let $\mathcal{C}_-$ be a finite nonempty set of actual monopole classes on $\bar{M}$. Suppose the following set of pairs is nonempty and define*

$$z_* := \frac{1}{e}\max\{\,|a\cdot b| : a \in \mathrm{Hull}(\mathcal{C}_+),\ a^2 \geq T_+,\ b \in \mathcal{C}_-,\ -b^2 \geq T_-\,\}.$$

Then, under FD2,

$$\mathcal{J}(g) + (1-t)_+ + (t-z_*)_+ \geq \varepsilon_{\mathrm{pair}}$$

for every smooth metric. Enlarging either supplied class set can only improve this inequality.

Proof. The maximum exists by compactness of the finite convex hull and finiteness of $\mathcal{C}_-$. Apply Theorem 2.1 to a maximizing pair. The last assertion follows from monotonicity of the maximum and of $(t-z_*)_+$. □

The resulting quantity uses the mixed intersection data, not only the separate class squares. It is a finite-dimensional cohomological optimization; no assertion is made here that it gives a complete characterization of all geometric polarizations. In particular it does not manufacture an opposite-orientation monopole class on a blowup where none has been supplied.

Corollary 2.3 (Einstein consequence)**.** *Under the preceding hypotheses, an Einstein metric must satisfy*

$$t \leq 1, \qquad \min\{t, z_*\} \leq 1 - \varepsilon_{\mathrm{pair}}.$$

If $z_* \geq t - \delta$, with $0 \leq \delta < \varepsilon_{\mathrm{pair}}$, then

$$3\tau \leq (1 - \varepsilon_{\mathrm{pair}} + \delta)e.$$

Proof. An Einstein metric is negative by the actual positive scalar budget and has $j = 0$. Equation (2.4) gives $t \leq 1$. For $t \leq 1$, $(1-t) + (t-z_*)_+ = 1 - \min\{t, z_*\}$. The conclusion follows. □

3 Geometric inputs: bundles and symplectic pairs

The surface-bundle classes in the supplied proof satisfy

$$a = x + y, \quad b = x - y, \quad x^2 = 0, \quad x \cdot y = e, \quad y^2 = 3\tau.$$

Thus $a^2 = T_+$, $-b^2 = T_-$, $|a \cdot b| = 3\tau$, so Theorem 2.1 recovers the existing surface-bundle inequality with the same convenient constant. This recovery is not a separate new solution of the Kodaira-slope question.

A symplectic pair is a pair of closed rank-two forms whose kernel foliations are complementary. The forms $\omega_1 + \omega_2$ and $\omega_1 - \omega_2$ induce opposite orientations [6]. Write the corresponding oriented plane-bundle splitting as

$$TM = E_1 \oplus E_2, \qquad x = e(E_1), \quad y = e(E_2).$$

The two first Chern classes are, up to an overall sign, $a = x + y$ and $b = x - y$, and

$$x \cdot y = e, \qquad x^2 + y^2 = 3\tau.$$

For example, $b_2^+(M), b_2^-(M) > 1$ ensure actual canonical-class supply through the usual symplectic monopole theorem. With that supply and $e > 0$, the new general inequality specializes to

$$\boxed{\mathcal{J}(g) + \left(1 - \frac{3\tau}{e}\right)_+ + \frac{2}{e}\big(\min\{x^2, y^2\}\big)_+ \geq \varepsilon_{\mathrm{pair}}.} \tag{3.1}$$

Indeed, since $x^2+y^2=3\tau\geq 0$, $(3\tau-|x^2-y^2|)_+=2(\min\{x^2,y^2\})_+$. The formula explicitly records the extra topological term; it does not assume that this term vanishes for every symplectic pair. If one Euler square is nonpositive, the surface-bundle form of the inequality follows without requiring either foliation to be a fibration. Classical constructions include symplectic pairs on manifolds that are not surface bundles [6]; the Euler squares and actual supply must be evaluated on a chosen such example before declaring a new individual application.

4 Sharp sectional pinching of the complete explicit model

This part is independent of FD2. Its input is the system's complete simply connected pair (N,g,Ω) with $N\cong\mathbb{R}^4$, $\mathrm{Ric}=-3g$, $d\Omega=0$, $|\Omega|^2=2$ and

$$|\nabla\Omega|^2=\frac{36(1-x^2)^3}{(1+3x^2)^3}>0,\qquad 1/3\leq x<1.$$

The construction, bolt extension and completeness belong to the source model [3, 4]. The curvature blocks computed there are

$$\mathcal{R}^-=-I,\qquad \mathcal{R}^+=\mathrm{diag}(r_1,r_2,r_3),\qquad D=1+3x^2, \tag{4.1}$$

where

$$r_1=-\frac{3(x-1)^2(21x^4+18x^3+12x^2-2x-1)}{D^3}, \tag{4.2}$$

$$r_2=-\frac{3(x+1)^2(21x^4-18x^3+12x^2+2x-1)}{D^3}, \tag{4.3}$$

$$r_3=\frac{9(x-1)^2(x+1)^2(5x^2-1)}{D^3}. \tag{4.4}$$

They obey $r_1+r_2+r_3=-3$.

Theorem 4.1 (Sharp global sectional range)**.** *For the source complete metric, every sectional curvature satisfies*

$$\boxed{-2<\sec_g\leq-\tfrac{1}{3}.}$$

The upper endpoint is attained at $x=1/\sqrt{3}$, and the infimum -2 is approached as $x\to 1$. Thus the complete strictly almost-Kähler example is uniformly negatively curved, with sharp global pinching ratio $1/6$ in this normalization.

Proof. The following rational factorizations determine all needed signs:

$$r_1-r_2=\frac{48x^3}{D^2}>0, \tag{4.5}$$

$$r_3-r_2=\frac{12(x+1)^2(3x-1)(3x^3-3x^2+x+1)}{D^3}\geq 0, \tag{4.6}$$

$$r_2'=\frac{72x(x-1)^2(x+1)(3x^2-3x-2)}{D^4}<0\qquad(1/3<x<1), \tag{4.7}$$

$$\tfrac{1}{3}-r_3=\frac{4(7-3x^2)(3x^2-1)^2}{3D^3}\geq 0. \tag{4.8}$$

The cubic in the second line is $3(x-1/3)^3+10/9>0$. Since $r_2(1/3)=-4/3$ and $\lim_{x\to1}r_2=-3$, we have $-3<r_2\leq-4/3$. For $z=x-1/3\geq 0$,

$$21x^4+18x^3+12x^2-2x-1=21z^4+46z^3+44z^2+\tfrac{136}{9}z+\tfrac{16}{27}>0,$$

so $r_1 \le 0$. It follows that

$$-3 < \min_i r_i = r_2, \qquad \max_i r_i \le 1/3.$$

A unit simple bivector has equal squared norms $1/2$ in its self-dual and anti-self-dual projections. Therefore its sectional curvature is

$$\sec(\sigma) = \tfrac{1}{2}\langle \mathcal{R}^+ u, u\rangle - \tfrac{1}{2}, \qquad |u| = 1.$$

Conversely every unit $u \in \Lambda^+$ occurs in such a bivector, by pairing it with any unit element of Λ^-. Hence the pointwise extrema are $(\min_i r_i - 1)/2$ and $(\max_i r_i - 1)/2$. The displayed bounds and the equality $r_3(1/\sqrt{3}) = 1/3$ prove the result. □

The strengthening here is sectional, not another verification of the Einstein equation. It shows that completeness, uniform negative sectional curvature and the Einstein almost-Kähler equations are simultaneously compatible with strict nonintegrability. This remains a noncompact statement.

5 No finite-volume quotient of the metric

The source already excludes compact quotients that preserve both g and Ω using the torsion amplitude. The next conclusion needs only the metric and applies without preserving Ω.

Theorem 5.1 (Metric quotients have infinite volume)**.** *Let Γ be any discrete group acting freely and properly by isometries of the complete metric in Theorem 4.1. Then*

$$\mathrm{Vol}(N/\Gamma) = \infty.$$

Proof. The smallest eigenvalue of the full curvature operator is $r_2(x)$, since $r_2 \le -4/3 < -1$ and $r_2 \le r_1, r_3$. Its strict monotonicity recovers x intrinsically. Every isometry therefore preserves x and the bolt $x = 1/3$.

On the regular set write $dr = \sqrt{G(x)}\,dx$. An isometry preserving r preserves its unit gradient, hence commutes with radial flow. In the product representation

$$N \setminus \text{bolt} = (0,\infty)_r \times \mathrm{SL}(2,\mathbb{R}),$$

it has the form $(r,p) \mapsto (r, \varphi(p))$, with φ independent of r. Thus the quotient of every regular level is the same manifold $Y = \mathrm{SL}(2,\mathbb{R})/\Gamma$ (with the action just obtained, not assumed to be left translation). The invariant orbit density $|\sigma_1 \wedge \sigma_2 \wedge \sigma_3|$ descends, because it is a positive constant multiple of the volume density of any fixed level. Its total mass $\mathfrak{v}$ is positive, possibly infinite.

The source metric has volume density

$$dV = \frac{64x(1+3x^2)}{3(1-x^2)^3}\,dx\,|\sigma_1 \wedge \sigma_2 \wedge \sigma_3|. \tag{5.1}$$

The bolt has measure zero. Tonelli's theorem and (5.1) give

$$\mathrm{Vol}(N/\Gamma) = \mathfrak{v}\int_{1/3}^{1} \frac{64x(1+3x^2)}{3(1-x^2)^3}\,dx = \infty,$$

since the integrand is asymptotic to $32/[3(1-x)^3]$. Freeness and properness ensure that Y contains an open set of positive quotient measure, so $\mathfrak{v}$ cannot vanish. □

Corollary 5.2 (An Einstein germ cannot be preserved in a finite-volume completion)**.** *No connected complete finite-volume Einstein four-manifold has a nonempty open subset locally isometric to the source metric. In particular, no closed Einstein metric can retain this metric exactly on an open patch while altering its complement.*

Proof. Einstein metrics are real analytic in harmonic coordinates [7, Theorem 5.2]. A local isometry between complete simply connected real-analytic Riemannian manifolds extends to a global isometry, by analytic continuation of local isometries [8, Chapter VI, Corollary 6.4]; compare [7, Theorem 5.4]. Apply this to $N \cong \mathbb{R}^4$ and the complete universal cover of the proposed Einstein manifold. Its universal cover is then isometric to N, so the manifold is an isometric quotient of N. Theorem 5.1 contradicts finite volume. □

This conclusion concerns preservation of an exact Einstein germ. It does not rule out changing the model everywhere, nor does it give a quantitative obstruction to approximate gluing. In particular it is not a proof of the closed Goldberg conjecture.

6 What has changed in the common research line

The first continuation replaces the exact bundle pairing by a quantitative mixed-pairing loss and allows convex-hull input on one side. Its dependence is

$$\text{actual monopole scalar bounds} + \text{FD2} + \text{same-metric Hodge pairing} \Longrightarrow (2.1).$$

It extends an established project implication but does not establish FD2 independently, and it has not supplied an opposite-orientation class for every nonminimal general-type surface.

The second continuation starts from the system's explicit complete metric, and obtains sharp sectional pinching and a metric-only obstruction to finite-volume quotients. It uses no monopole endpoint estimate. The original Fino construction remains the system's contribution; the new content here is the global curvature range and the consequence of the intrinsic radial eigenvalue.

No smooth non-Bach-flat absolute P-minimizer is constructed in this note. General O10 and the sharp unrestricted P infima also retain their prior status. The results above have their stated narrower completion criteria and proofs.

References

[1] Qiushi project, *O10_All_Finite_Density.md*, finite-density endpoint statement; the present note uses only its fixed $B = 2$ instance.

[2] Qiushi continuation, *Line1_Two_Orientation_Global_Gap.md*, 7 October 2026, Sections 1–6.

[3] System A01 construction and companion continuation, *Q4_Complete_Example_and_Fino_Question.tex*, 8 October 2026, Sections 1–4.

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[5] C. LeBrun, *Four-Manifolds, Curvature Bounds, and Convex Geometry*, arXiv:math/0611450, Sections 3–4. `https://arxiv.org/abs/math/0611450`.

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